

Q.6 1**(a) 1Points of Presence (PoP) in Amazon CloudFront**

[2+2 marks]

Question

Describe the functionality of Points of Presence (PoP) in Amazon CloudFront, and briefly describe its components.

A **Point of Presence (PoP)** in Amazon Web Services CloudFront is an **edge location** where cached copies of website content are stored.

Its main function is to deliver content (images, videos, web pages, etc.) to users from the **nearest server**, which reduces **latency** and improves **loading speed**.

Example**Example:**

If a website server is in the USA and a user accesses it from Bangladesh, CloudFront delivers the content from the nearest PoP in Asia instead of the USA, making the website faster.

Components of CloudFront PoP**1. Edge Servers**

Store cached content and send it to nearby users quickly.

2. Cache Storage

Keeps temporary copies of files such as images, CSS, videos, and web pages.

3. Routing Network

Directs user requests to the nearest and best-performing PoP.

4. Security Features

Provides HTTPS, DDoS protection, and traffic filtering for secure content delivery.

(b) 1Eight AWS Services

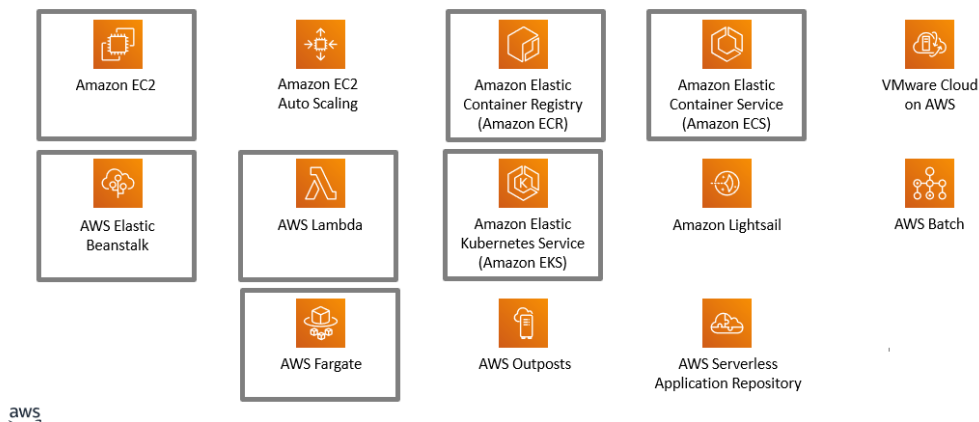
[2 marks]

Question

Write the names of eight services in AWS.

AWS compute services

Amazon Web Services (AWS) offers many compute services. This module will discuss the highlighted services.



Eight AWS Compute Services

1. Amazon EC2
2. Amazon EC2 Auto Scaling
3. Amazon Elastic Container Registry (ECR)
4. Amazon Elastic Container Service (ECS)
5. AWS Elastic Beanstalk
6. AWS Lambda
7. Amazon Elastic Kubernetes Service (EKS)
8. Amazon Lightsail

(c) 1Auto Scaling Group

[2+2 marks]

Question

What is an Auto Scaling Group? Describe Dynamic Scaling Implementation.

An **Auto Scaling Group (ASG)** in Amazon Web Services is a collection of EC2 instances that are managed together for **automatic scaling** and **high availability**.

Example

Example:

If an e-commerce website gets many users during a sale, the Auto Scaling Group automatically launches more EC2 instances to handle the extra traffic.

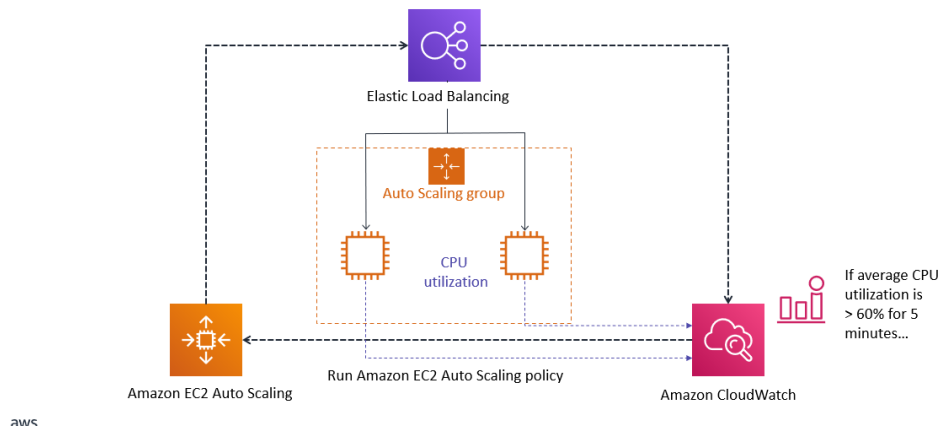
Dynamic Scaling Implementation

Dynamic scaling automatically increases or decreases EC2 instances based on system performance.

Simple Process:

1. Amazon CloudWatch monitors CPU usage of EC2 instances.
2. If CPU usage becomes high (for example above 60%), Auto Scaling automatically adds new EC2 instances.
3. Elastic Load Balancer distributes traffic among all instances.

Implementing dynamic scaling



Q.5 1

(a) 1Network ACL Inbound/Outbound Rules

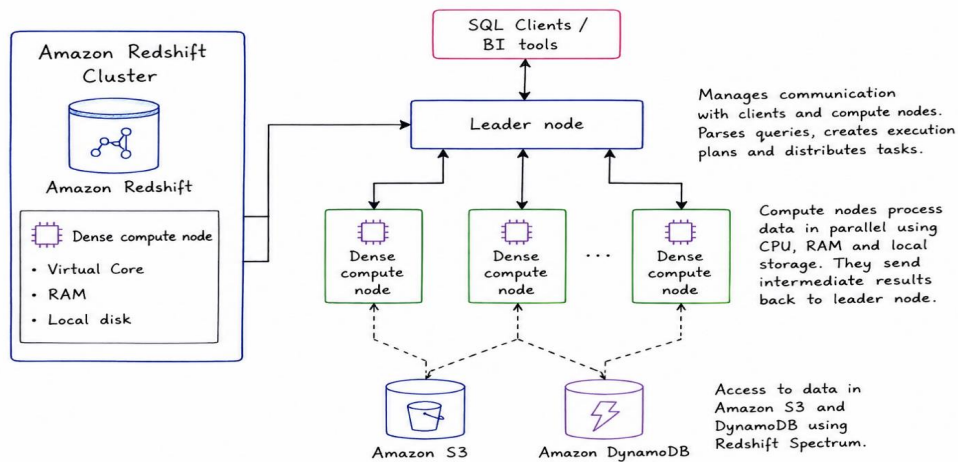
Note: There are some corrections in this section — *wait for the next update.*

Inbound					
Rule	Type	Protocol	Port Range	Source	Allow/Deny
100	HTTPS	TCP	443	0.0.0.0/0	ALLOW
120	SSH	TCP	22	192.0.2.0/24	ALLOW
*	All IPv4 traffic	All	All	0.0.0.0/0	DENY
Outbound					
Rule	Type	Protocol	Port Range	Destination	Allow/Deny
100	HTTPS	TCP	443	0.0.0.0/0	ALLOW
120	SSH	TCP	22	192.0.2.0/24	ALLOW
*	All IPv4 traffic	All	All	0.0.0.0/0	DENY

MVC

(b) 1S3-Triggered Lambda Thumbnail Process

1. User uploads image (.jpg/.png) to the source S3 bucket.
2. Amazon S3 detects the object-created event after the image upload.
3. S3 triggers the Lambda function and sends the event data (bucket name and object key).
4. Lambda assumes the execution role to get required permissions through the access policy.
5. Lambda reads the uploaded image, creates a thumbnail using graphics libraries, and saves the thumbnail to the target S3 bucket.

(c) 1Parallel Processing Architecture of Amazon RedshiftParallel Processing Architecture of Amazon Redshift

Q.4 1**(a) 1Nine Key Decisions When Launching an EC2 Instance**

[5 marks]

Question

What are the nine key decisions to make when you create an EC2 instance using the AWS Management Console Launch Instance Wizard? Briefly describe each decision step.

Nine Key Decisions in the AWS EC2 Launch Instance Wizard

- 1. AMI (Amazon Machine Image)**
Choose the operating system and pre-installed software for the EC2 instance, such as Linux or Windows.
- 2. Instance Type**
Select the hardware configuration of the instance, including CPU, RAM, and storage capacity.
- 3. Network Settings**
Configure the VPC, subnet, and public IP settings to control network connectivity.
- 4. IAM Role**
Attach an IAM role to grant the EC2 instance permission to access AWS services securely.
- 5. User Data**
Add startup scripts or commands that automatically run when the instance launches.
- 6. Storage Options**
Configure EBS volumes or storage size that will be attached to the instance.
- 7. Tags**
Add labels (key-value pairs) to organize and identify EC2 resources easily.
- 8. Security Group**
Define inbound and outbound firewall rules to control traffic access to the instance.
- 9. Key Pair**
Select or create a key pair for secure login access to the EC2 instance using SSH or RDP.

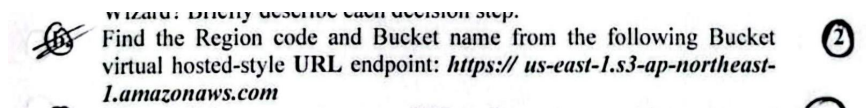
(b) 1S3 Bucket URL — Region & Bucket Name

[2 marks]

Question

Find the Region code and Bucket name from the following Bucket virtual hosted-style URL endpoint:

<https://us-east-1.s3.ap-northeast-1.amazonaws.com>



Bucket Name: us-east-1

Region Code: ap-northeast-1

For more information:

Bucket path-style URL endpoint:

<https://s3.ap-northeast-1.amazonaws.com/bucket-name>

Bucket virtual hosted-style URL endpoint:

<https://bucket-name.s3.ap-northeast-1.amazonaws.com>

(c) 1Amazon EBS Volume Types

[2+1 marks]

Question

Differentiate among Amazon EBS volume types with respect to maximum volume size, maximum IOPS/Volume, and maximum throughput/Volume. Write down the EBS features.

Bucket path-style URL endpoint:

<https://s3.ap-northeast-1.amazonaws.com/bucket-name>

Region code

Bucket name

Bucket virtual hosted-style URL endpoint:

<https://bucket-name.s3.ap-northeast-1.amazonaws.com>

Bucket name

Region code

		Solid State Drives (SSD)		Hard Disk Drives (HDD)	
		General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
Maximum Size	Volume	16 TiB	16 TiB	16 TiB	16 TiB
Maximum IOPS/Volume		16,000	64,000	500	250
Maximum Throughput/Vol		250 MiB/s	1,000 MiB/s	500 MiB/s	250 MiB/s

EBS Features

- Provides **persistent block storage** for EC2 instances.
- Supports **high availability and durability**.
- Allows **snapshot backup** using Amazon S3.
- Volumes can be **attached, detached, and resized** easily.

Q.3 1**(a) 1EC2 Instance Lifecycle****Question**

Draw the EC2 Instance lifecycle.

Amazon EBS volume types

	Solid State Drives (SSD)		Hard Disk Drives (HDD)	
	General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
Maximum Volume Size	16 TiB	16 TiB	16 TiB	16 TiB
Maximum IOPS/Volume	16,000	64,000	500	250
Maximum Throughput/Volume	250 MiB/s	1,000 MiB/s	500 MiB/s	250 MiB/s

(b) 1Factors Considered When Designing Data Centers

[4 marks]

Question

What factors are considered by cloud service providers when designing data centers?

Key Data Center Design Factors**1. Location Safety**

Data centers are built in safe places to avoid natural disasters.

2. Backup Systems

Extra power, network, and connectivity systems are used to avoid service failure.

3. High Security

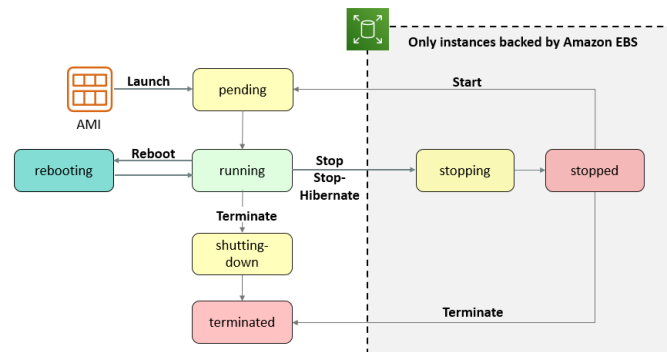
Access to data centers is restricted and locations are kept secret.

4. High Availability

Data and services are backed up across multiple locations to keep services running during failures.

(c) 1Four Pillars of Cost Optimization

Amazon EC2 instance lifecycle



Four Pillars of Cost Optimization

1. Right-Sizing

Selecting the correct instance type and size based on workload requirements to avoid wasting resources.

2. Elasticity

Automatically scaling resources up or down according to demand to reduce idle capacity.

3. Optimal Pricing Model

Choosing suitable pricing options such as On-Demand, Reserved, or Spot Instances based on usage patterns.

4. Storage Optimization

Using appropriate and low-cost storage options while removing unused storage to reduce expenses.